

Shamala Chandrappa

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SUMMARY

*Results-driven **AI Engineer** with a **Master's in Data Science** and over **5 years** of experience across **finance**, **Sales analytics**, and **healthcare** domains. Actively seeking **challenging opportunities** to build **impactful**, **fault-tolerant**, and **customer-focused software solutions**.*

CERTIFICATIONS & AWARDS

- **IBM Python for Data Science, AI, and Development Professional Certificate**
- **Databases and SQL for Data Science with Python**
- **Google Cloud Fundamentals: Core Infrastructure By Google Cloud**
- **Snowflake Data Engineering Professional Certificate - Inprogress**

Associated with FIS Global

- Received **Best Employee of the Quarter Award**, recognized multiple times for outstanding performance, cross-functional collaboration, and timely delivery of high-impact analytics solutions.
- Received **Exceptional Female Employee Award** and multiple **"Shout-out"** for excellence in project delivery, leading cross-border payment modernization, and taking initiatives, enhancing platform stability.

PROFESSIONAL EXPERIENCE

San José State University Research Foundation

Research Assistant (Data Engineer - AI & LLM Safety)

San Jose, CA

Jan 2025 - Sep 2025

*Collaborated with **Dr. Simon Shim (SJSU)** and **Dr. Kichang Lee (Harvard Medical School)** on research applying **Generative AI** and predictive **ML** to healthcare data, focusing on safe, interpretable models for clinical decision support and risk prediction in cardiac surgery patients.*

Project 1: Delirium Prediction for Cardiac Surgery Patients

*Partnered with **Harvard Medical School's cardiac ICU** team to predict postoperative delirium vs non-delirium outcomes using multimodal EHR, biosensor, and lab data collected from cardiac-surgery patients.*

- Worked with a dataset of **1,700** proteins, including bio-proteins and biochemical indicators measured via sensors at **24 hr**, **36 hr**, and **72 hr** pre- and post-surgery, capturing the physiological signatures leading up to and following surgery
- Developed data ingestion and transformation pipelines using **Python** (Pandas, **PySpark**, **SQLAlchemy**) for scalable ETL/ELT workflows.
- Built reproducible training and evaluation workflows using **MLflow**, **Weights & Biases**, and **GitHub Actions** for automated **CI/CD**, ensuring model lineage and version control.
- Designed privacy-preserving **data pipelines** with data anonymization, IAM-based access controls, and audit logging to meet enterprise privacy requirements.
- Fine-tuned domain-adapted **LLMs** and retrieval-augmented generation (**RAG**) **pipelines** with vector databases (**Pinecone**, **FAISS**) to improve task accuracy and reduce hallucinations.
- Implemented data labeling, schema enforcement, and **PII detection** with Great Expectations and pandas-profiling, improving data reliability and compliance
- Developed explainability dashboards using **SHAP**, **LIME**, and Streamlit for feature importance visualization and drift detection
- Leveraged **Python** in building, training, and deploying machine learning models using leading frameworks such as **scikit-learn**, **TensorFlow**, and **PyTorch**
- Built and validated Random Forest, XGBoost, and LSTM models to detect early signs of delirium risk, **achieving ≈82 % accuracy and 0.79 F1 score**

- Engineered **300 +** clinical and temporal features from vitals, lab trends, and protein-expression patterns; applied SHAP interpretability to identify key risk factors such as ICU stay length, pre-op medications, and inflammatory protein markers.
- Maintained **HIPAA-aligned data pipelines** and audit logs, ensuring privacy, compliance, and reproducibility across institutions.

Project 2: Generative AI for Clinical Question Answering

- Developed domain-adapted **LLMs (Qwen, Mistral, Phi-3)** fine-tuned with **QLoRA** on anonymized **EHR datasets**, improving clinical question-answering 32% accuracy.
- Integrated **LLMs** with retrieval-augmented generation pipelines (LangChain + Pinecone) to summarize patient histories and surface relevant clinical notes, cutting physician decision turnaround by **40%**
- Deployed **FastAPI** microservices on **AWS Bedrock** using **Trainium/Inferentia** for low-latency inference, with continuous evaluation harnesses (**BLEU, ROUGE, BERTScore**) to monitor accuracy and reduce hallucinations

Result: Delivered a trustworthy **AI assistant** and predictive-risk platform that combined biosensor **data, molecular signals**, and **EHR** context to accelerate decision-making for cardiac surgeons, demonstrating how safe, auditable AI can enhance patient outcomes in **HIPAA-regulated** settings.

Peoplelens.ai

Remote, USA

Data and AI Engineer Intern

Jun 2024 - Aug 2024

*Delivered data-driven insights as part of the **Data & Analytics team**, supporting Go-to-Market and Revenue Operations teams to enhance service delivery and drive operational efficiency.*

Project 1: Designed governed analytics pipelines on **GCP** to unify marketing and sales data, improving reliability and reducing ad-hoc reporting workload

- Built streaming and batch pipelines using **GCP Dataflow, Pub/Sub, and Composer** to consolidate data from 10+ sources (**Salesforce, Google Calendar, Highspot**) into BigQuery and Snowflake.
- Improved data accessibility by **70%** and reduced **ad-hoc data requests** by half through a well-defined lakehouse model

Project 2: Data Quality & Governance

- Implemented **PII-aware ELT validation** and **schema enforcement**, ensuring GDPR alignment and cutting data issues by **40%**
- Enabled unified analytics for **GTM teams**, supporting faster decision-making on campaign performance and customer conversions

Other Duties

- Conducted **ad-hoc revenue analysis** using **Python** and **Excel** to investigate **data discrepancies**, identifying **root causes** and delivering corrected insights that empowered sales leaders to better track territory performance and revenue trends
- Collaborated with cross-functional teams to develop and deploy a regression model in **Python** that reduced forecast error in resource utilization by **20%**, optimizing schedule tracking by **40%**
- Implemented data quality checks on the data model and database, improving reporting accuracy by **30%** and reducing rework through schema validation and cross-source reconciliation
- Designed and implemented an error removal tool on Excel to reduce the time spent on data transfer by **25%**
- Leveraged **JIRA** to track project progress & maintained extensive documentation on Confluence pages
- Designed an internal **UAT process** for reports and provided training to key users, leading to **30%** fewer change requests in the reports and a higher number of active report users

FIS Global

Bengaluru, India

Senior Software Engineer - Development, ML & Data Platforms

Jul 2019 - Jul 2023

Client Name: Bandhan Bank

Cloud Migration and Reliability Engineering

*The legacy **on-prem** payment system was slow and **fragile**, often struggling with growing transaction volumes. To fix this, I led the migration to **AWS** and re-architected services into **modular Java** and **Spring Boot** microservices.*

- Built a **multi-region, fault-tolerant architecture** using **AWS Global Accelerator**, **Route 53**, and **ALB**, improving uptime and disaster recovery
- Automated infrastructure provisioning with Terraform and CloudFormation, and set up **CI/CD pipelines (CodePipeline + CodeBuild)**, cutting release cycles from weeks to days
- Rewrote key payment workflows in **Java/Spring Boot**, integrating **PostgreSQL** and **Redis** for faster, more reliable transaction processing
- Implemented detailed logging, metrics, and alerting through **CloudWatch** and Grafana, giving teams full observability into every stage of the payment flow

Result: Reduced manual intervention during outages by **60%**, improved reconciliation accuracy, and created a scalable foundation that now supports **10M+** transactions per day.

Client Name: Global Payments Platform

Real-Time Fraud Detection & Streaming Analytics

*Architected an **ML-driven fraud-detection** system using **AWS SageMaker**, **Kinesis**, and **EMR (Spark)** that processed live payment data across regions.*

- Built high-throughput ingestion pipelines (handling **500M+** transactions per day), feeding into a centralized **S3 data lake** and **Redshift BI layer**
- Designed end-to-end **SageMaker pipelines** with drift detection (**CloudWatch + Lambda**) and automated retraining
- Achieved a **40% reduction** in false positives while maintaining strict PCI compliance and real-time detection capabilities

Client Name: First Citizens Bank

Project 1: API Modernization

*Worked on modernizing the **bank's** transaction platform by building gRPC-based **microservices** for inter-service communication, improving performance, and developer productivity.*

- Developed secure, low-latency APIs in **Java + gRPC**, integrated with **API Gateway** and **PostgreSQL** for transactional persistence
- Expert in crafting high-performance, complex **SQL queries** for data extraction, transformation, and analysis across large-scale, relational databases, optimizing queries for speed and efficiency to drive actionable business insights
- Applied strict **PCI DSS standards**, with **TLS 1.2 encryption** and **CSRF/XSS** mitigation to safeguard all payment operations.
- Enhanced observability with **Grafana dashboards** and **Prometheus alerts**, helping engineers detect failures in real time.
- Mentored **12+** engineers, conducted **50+ code reviews**, and enforced test-driven modular service patterns, improving release consistency.

Result: Improved API latency by **45%**, cut manual debugging by **70%**, and established a uniform gRPC contract across all payment services.

Project 2: Data Lake & Automated Reconciliation Pipelines

Built an AWS-based data lake and automated reconciliation system to make financial operations more reliable and insights easier to access.

- **Automated daily reconciliation:** Designed **Dagster-based workflows** on **AWS** to match and verify payment transactions from the CardPointe Funding API, replacing manual checks and improving data accuracy
- **Enabled real-time monitoring:** Integrated **S3**, **Lambda**, **Athena**, and **CloudWatch** to automatically detect failed or missing settlements, send alerts, and keep teams informed with daily summary reports.
- **Boosted performance and reliability:** Optimized data-lake design and added disaster-recovery workflows, improving query speed by **~40 %** and overall reconciliation time by **35 %**.

Excel Parser System - RightSkale.ai (Open Source AI Project)

Architected an end-to-end AI-powered Excel Parser System that converts raw Excel data into actionable business insights using semantic retrieval and LLM-driven contextual reasoning.

- Developed a **modular pipeline integrating React/Tailwind UI with Supabase (Postgres + pgvector) backend** and serverless **Edge Functions for low-latency AI inference**.
- Implemented embedding generation (**gte-small, OpenAI embeddings**), vector-search-driven **GraphRAG** for intelligent natural-language queries on spreadsheet data.
- Delivered a real-time **sportsbook BI dashboard** visualizing **KPIs-Revenue, Profit Margins, Win Rate, Fraud Alerts**, and **Risk Exposure**-with AI-powered anomaly detection and dynamic odds pricing.
- Improved operational performance: **+15-25 % profit margin, 80% fraud loss reduction**, and 60% faster risk response.

FHIRBAND: AI-Powered Wearable for First Responders

TreeHacks 2025, Stanford University, CA

- Created an **AI-driven real-time health monitoring system** for first responders with a chatbot for instant health insights, integrating **Gemini Flash 2.0** for diagnostics and risk assessment.
- Implemented a **data pipeline (Firebase GCP BigQuery)** with **Cloud Functions** for streaming, analysis.
- Established a **secure, FHIR-compliant architecture** for medical record systems, ensuring interoperability, improving squad-level analytics dashboard accuracy by 15%.

Financial Benchmarking & Anomaly Detection Platform

- Architected a **Snowflake-based financial data platform** integrating SEC filings via GCS external stages into structured schemas for benchmarking and anomaly detection.
- **Web-scraped SEC filings from sec.gov**, extracted key fields (financial statements, revenue, expenses, headcount, disclosures), and performed **data transformation and standardization**.
- Applied **Cortex ML functions** on General Ledger data to detect anomalies, with automated alerts, **LLM-powered RAG explanations** using **Claude + Agentic AI**.

AI Mock Interviewer | Project Lead, Applied Data Science Department

Developed an **AI-powered interview assistant** that provides **24/7 personalized feedback and scoring** using **AWS (Lambda, S3, RDS, SageMaker)**, **Spark**, and **Pinecone DB**.

- Designed and implemented the end-to-end system architecture, including data ingestion, processing, and **retrieval-augmented generation (RAG)** with **LLMs** for dynamic question generation.
- Built an evaluation engine leveraging **AWS Transcribe** and **RDS** to analyze responses, ensuring scalability, automation (**CI/CD with AWS Code Pipeline**), and real-time monitoring via **CloudWatch & X-Ray**.

Google Play Store Apps Analysis (Role: ML Backend Contributor)

- Employed **machine learning** to evaluate **app market trends** and **user preferences**, demonstrated proficiency in delivering insights and fostering data-driven decision-making.
- **Tech Stack:** Google Play Scraper, AWS SageMaker, EC2, Lambda, Glue, Crawler, ETL Jobs, VPC, Redshift, Apache Air Flow, and Power BI

SKILLS

Programming Languages: Python (NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch, Flask), SQL, Java, C++

Cloud Platforms: AWS (Lambda, API Gateway, S3, SQS, SNS, EC2, RDS, VPC, CloudFront, IAM, CloudWatch, CDK, Bedrock), GCP (BigQuery, Dataflow, GCS, Composer), Azure (ADF, Synapse, Databricks)

Data Engineering & Big Data: Airflow, Kafka, Spark, Flink, Glue, EMR, Hadoop, PySpark, Hive, Redshift, MapReduce, Cassandra

Databases: Snowflake, Databricks, PostgreSQL, MySQL, SQLite, MongoDB, DynamoDB, Neo4j, Firebase, Redis, NoSQL

AI & Machine Learning: Transformers, LLMs (fine-tuning & integration), Pinecone, LangChain, Supervised ML (Regression, Decision Trees, Random Forest, XGBoost, KNN, LDA), Unsupervised ML (K-Means, PCA), NLP, Statistical Analysis, A/B Testing

DevOps & Infrastructure: Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Git, Linux, Serverless, Cloud-native Deployments, REST, gRPC, Spring Boot, Multithreading

Observability & Testing: Unit Testing, Performance Tuning, System Profiling, Fault Injection, ELK Stack

Visualization & BI: Tableau, Power BI, QuickSight, Metabase, Excel, PowerPoint

Tools & Frameworks: Django, GraphQL, Postman, JIRA, Confluence

EDUCATION

San Jose State University

MS in Data Science

San Jose, California, USA

Aug 2023 - May 2025

CMR Institute of Technology - Visvesvaraya Technological University (VTU)

B.E. in Computer Science and Engineering

Bengaluru, India

Aug 2015 - May 2019